Toward Multi-SLO-Aware Scheduling for Heterogeneous AI Accelerators on Edge Devices

Agenda

- Motivation and Research Challenges.
- Proposed Approach
  1. Research Part #1 – Multi-SLO heterogeneous scheduling
  2. Research Part #2 – Model-aware dynamic inference adaptation
  3. Research Part #3 – Self-Calibrating Multi-SLO Adaptation
  4. Evaluation Plan
- Timeline, Summary, and Contributions

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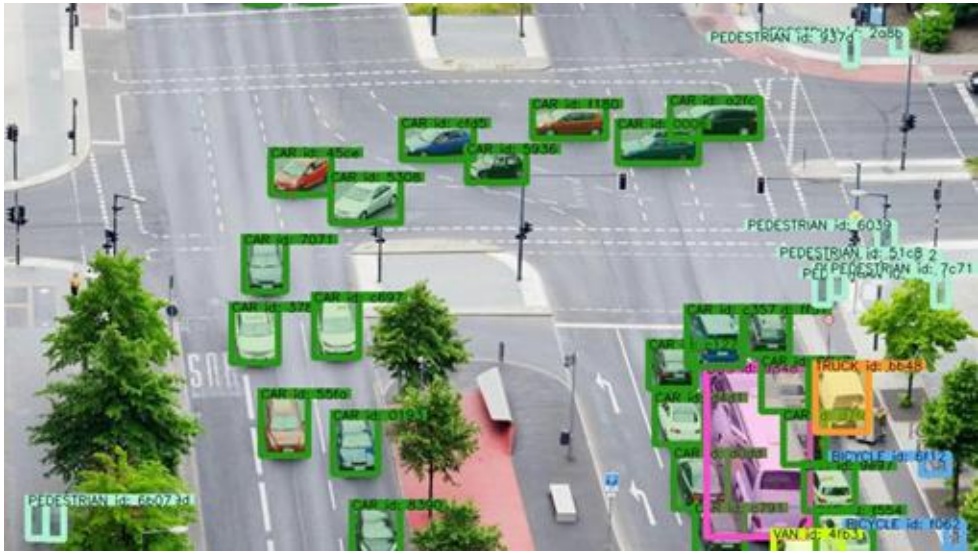
Research Motivation

- Real-Time Smart Traffic Camera
  - ❌ High latency → pedestrian detected too late → safety risk
  - ❌ Accuracy drop → false negatives → missed violations
  - ❌ Energy spike → thermal throttling → system slowdown



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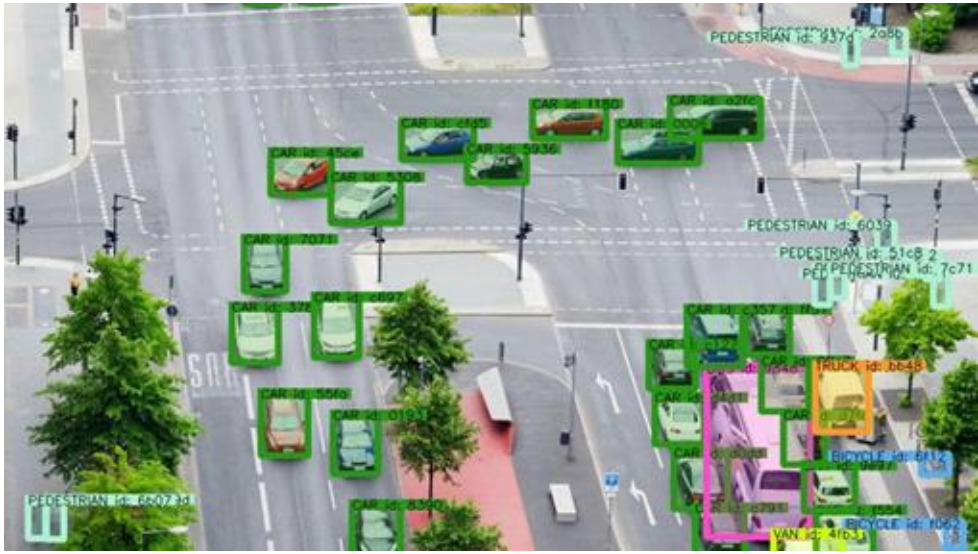
SLOs Matter

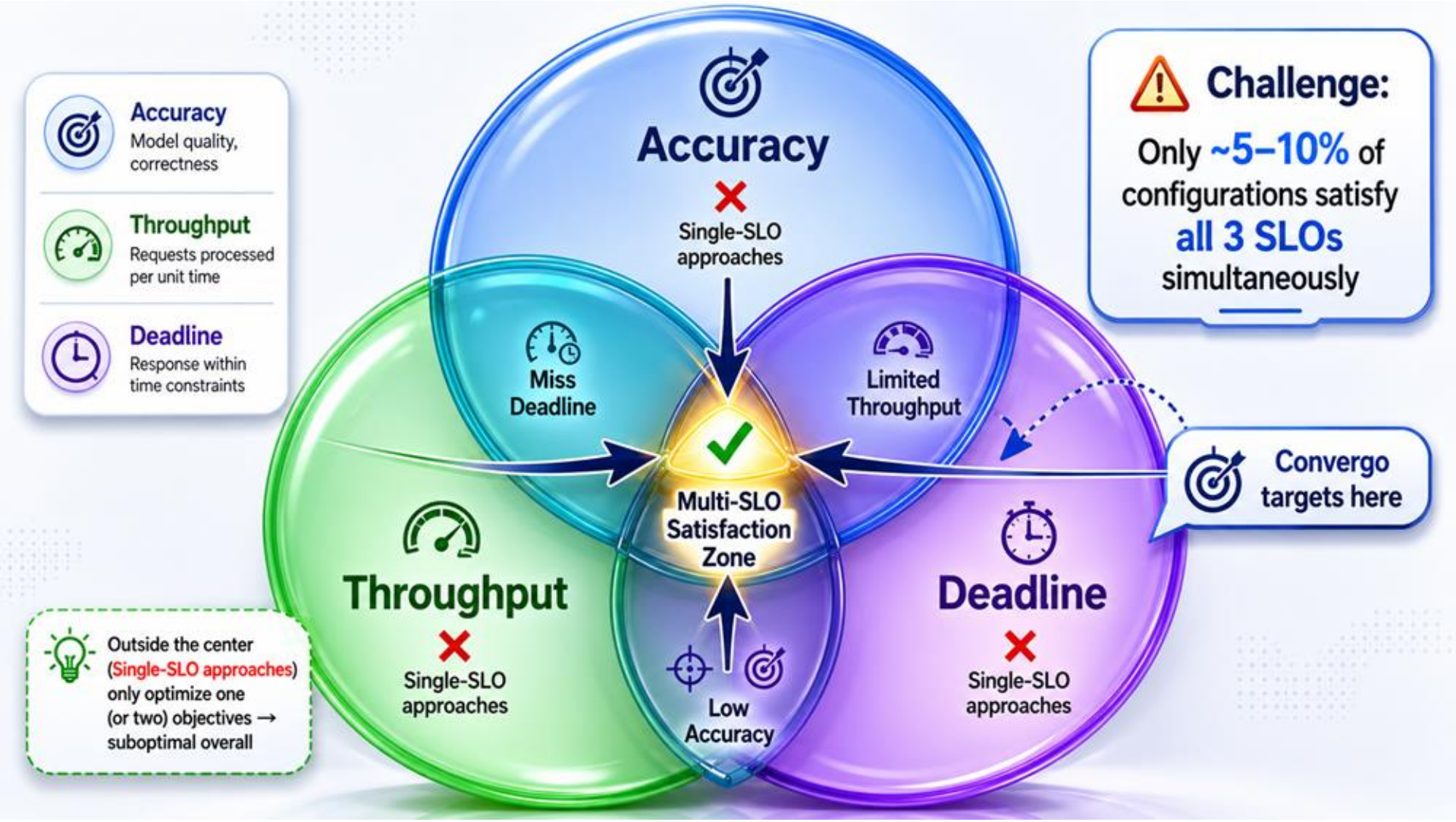


Research Challenges

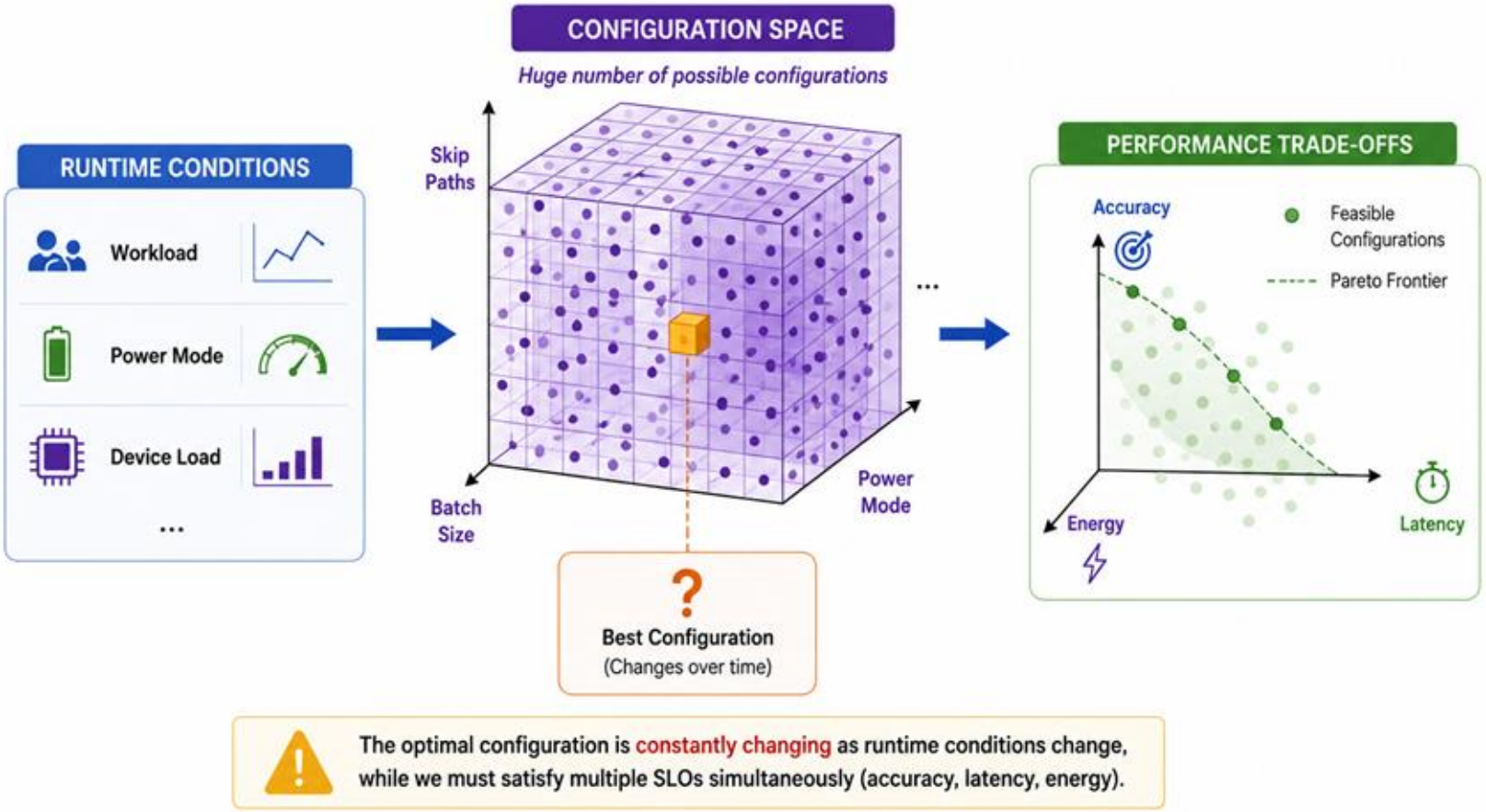
Current IoT-edge Systems:

- ❑ Multi-SLO Heterogeneous Scheduling
- ❑ Model-Aware Dynamic Inference Adaptation
- ❑ Elastic Model and Runtime Co-Design

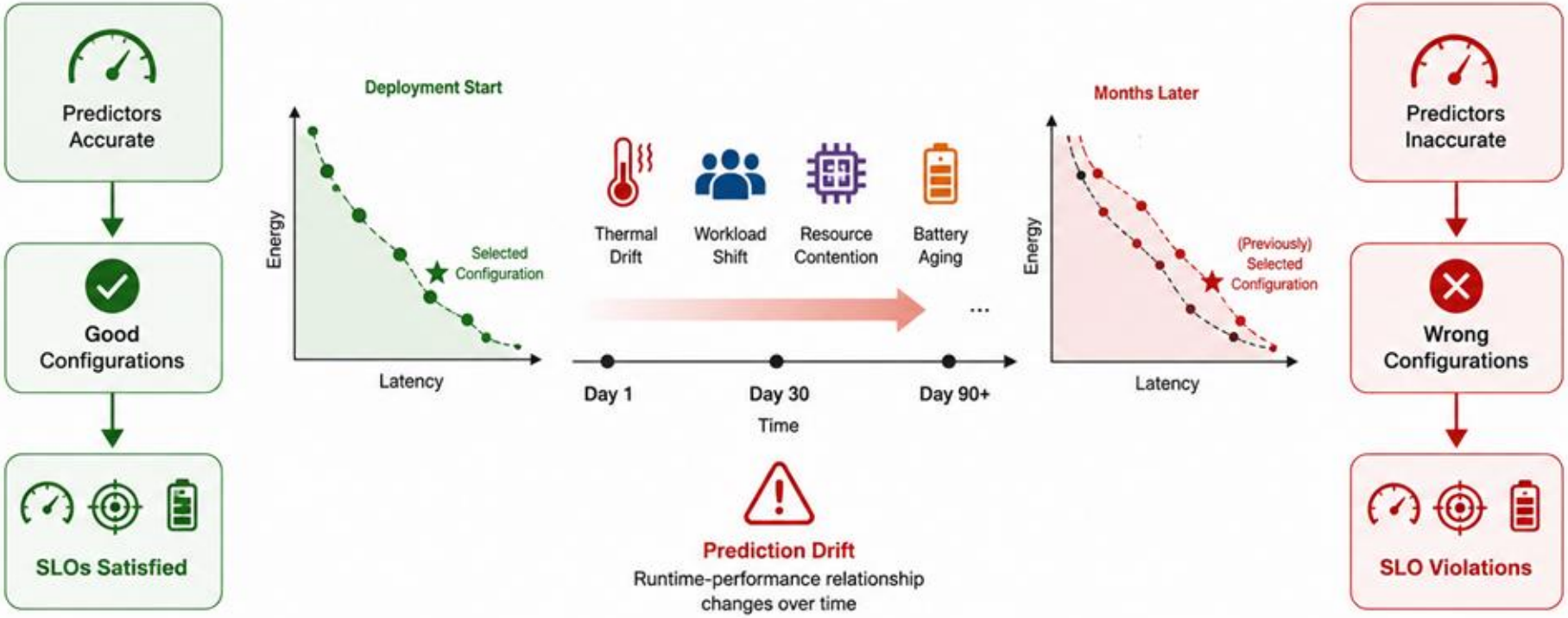
Challenge #1:
Multi-SLO Heterogeneous Scheduling



Challenge #2:
Model-Aware Dynamic Inference Adaptation



Challenge #3:
Self-Calibrating Multi-SLO Adaptation



Problem Statement

- How can edge AI systems continuously satisfy multiple service-level objectives despite dynamic workloads, evolving device conditions, and prediction drift in heterogeneous resource-constrained environments?

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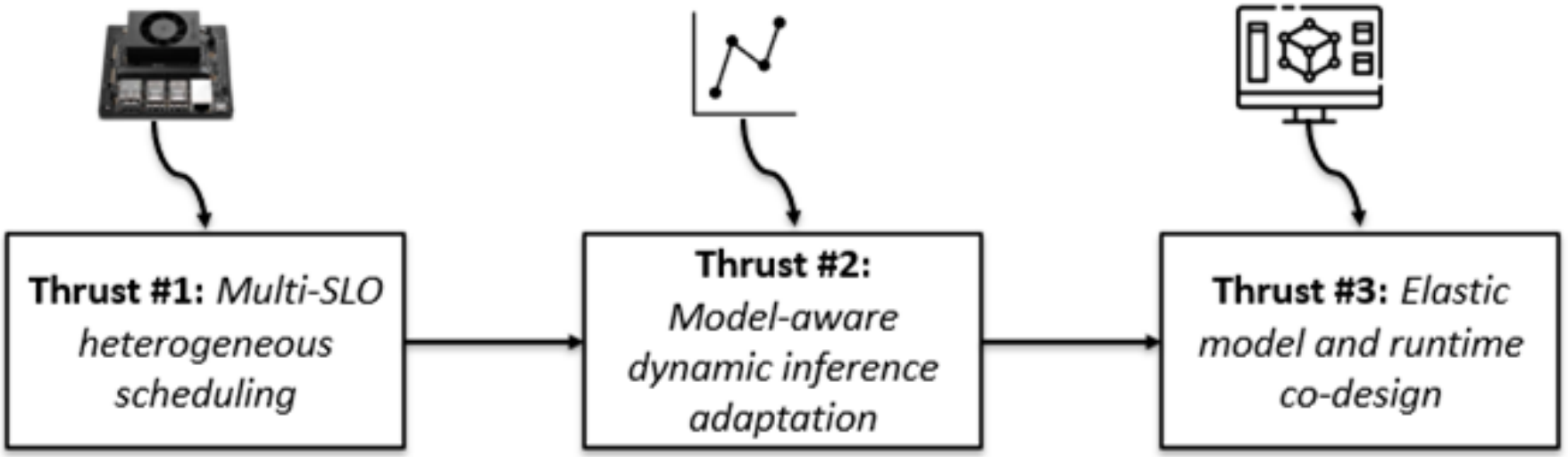
Proposed Research Overview



Research Thrust #1:

Multi-SLO heterogeneous scheduling

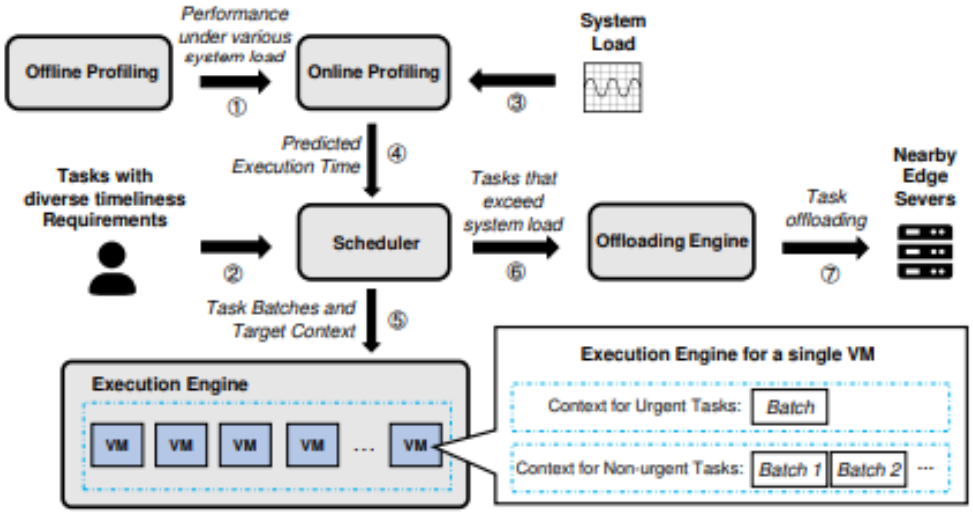
Research Thrust #1

- Challenge
  - Simultaneously Meeting Multiple SLOs (Service Level Objectives) in Edge AI Scheduling

- Research goal
  - Establish the foundational system-level orchestration layer that intelligently coordinates heterogeneous accelerators and static models to maximize multi-SLO satisfaction, while revealing the fundamental efficiency ceiling imposed by the black-box model paradigm.

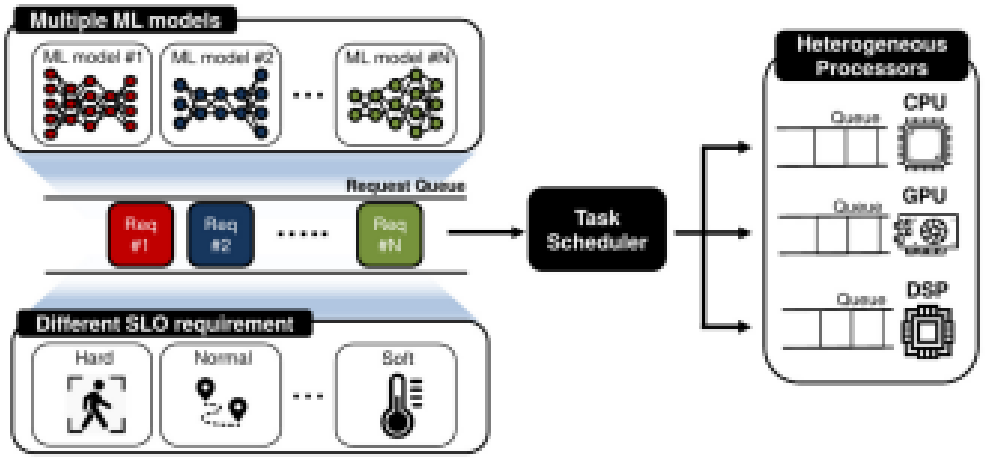
Research Thrust #1 -- Related Work

- Park, Seonyeong, et al. (IEEE Access '21)
  - Kalmia is a scheduling framework for edge servers that manages heterogeneous DNN inference tasks while meeting QoS requirements by co-locating different types of workloads (latency-critical and best-effort) on GPUs and CPUs.



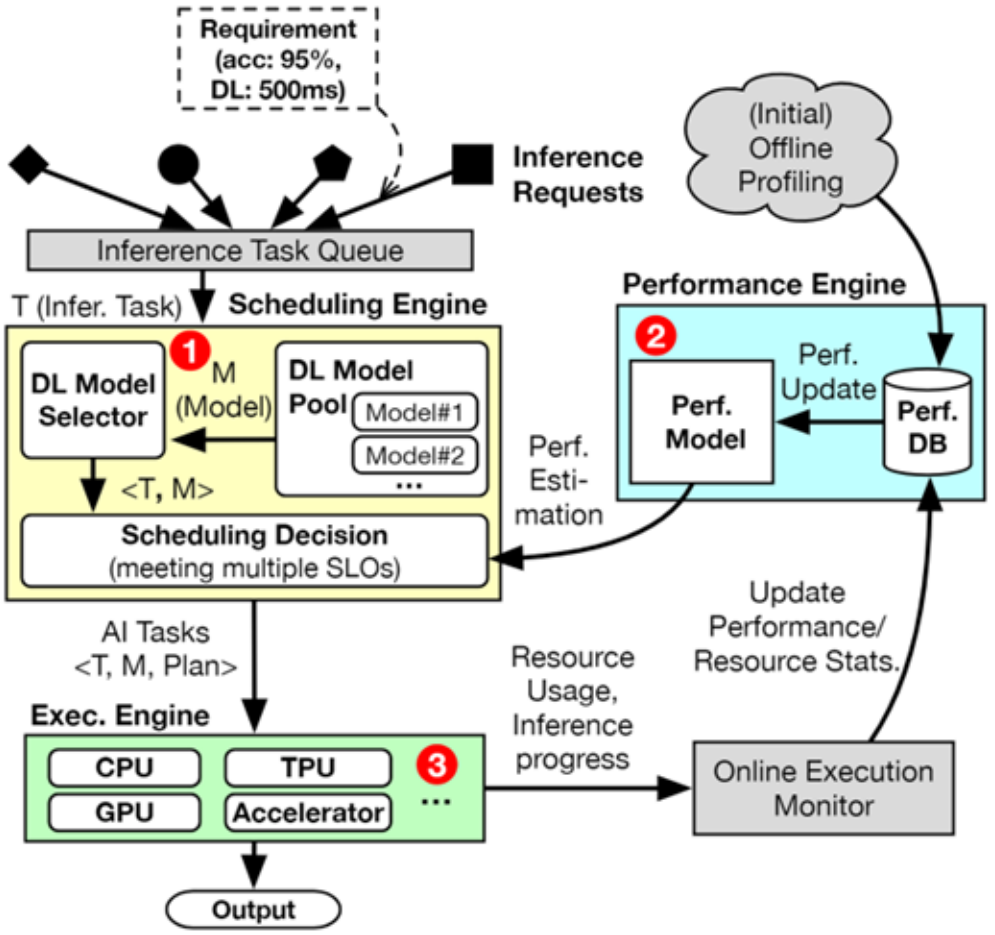
Research Thrust #1 -- Related Work

- Seo, Wonik, et al. (ACM TACO ’21)
  - This paper presents an SLO-aware scheduler for DNN inference on heterogeneous edge platforms (CPUs, GPUs, and accelerators) that dynamically assigns inference tasks to different processors to meet service level objectives (SLOs). The scheduler uses runtime profiling and prediction models to estimate execution times and makes scheduling decisions that maximize resource utilization while satisfying latency requirements.



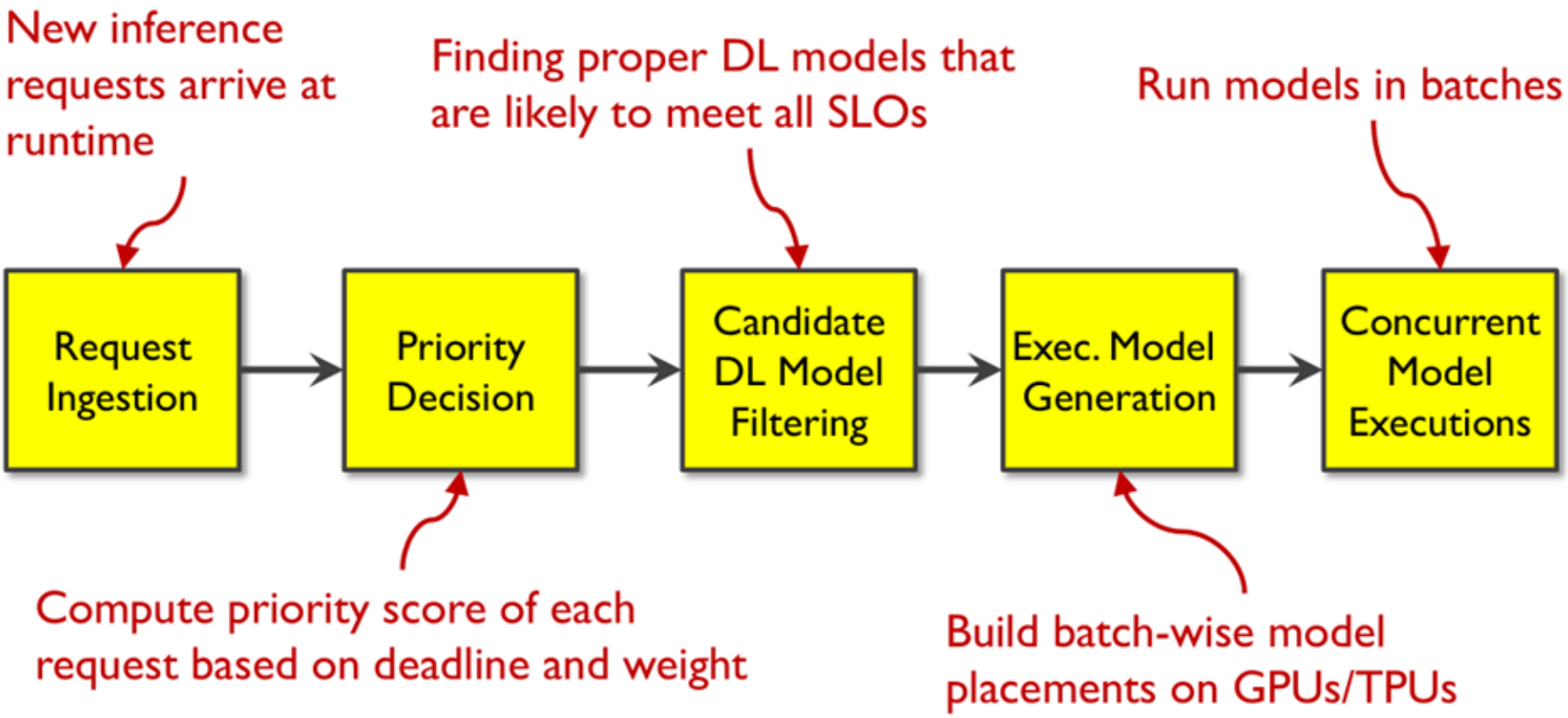
Research Thrust #1

1. Scheduling Engine
   - Select best models to meet SLOs
2. Performance Engine
   - Profiles and estimate model perf.
3. Execution Engine
   - Runs models on AI accelerators



Research Thrust #1

- Scheduling Overview



Research Thrust #1

- User Requests and Priority
  - Convergo handles multiple concurrent inference tasks with dynamic priority scheduling.
- Each user request (R_i) includes multiple SLOs:
  - Throughput (Th_i), Accuracy (Acc_i), Deadline (D_i)
- All requests are placed in a **priority queue**

Deadline Current Time Arrival Time

$$Priority_i = \frac{D_i - (t_{curr} - t_{arr,R_i})}{w_i}$$

w_i ← User assigned weight (1 -- 10)

- Priority scores are dynamically updates over time

Research Thrust #1

- After prioritization, Convergo checks 3 constraints to find candidate DL models.
  - Constraint #1: Throughput $\sum_{j=1}^{n} \mathcal{M}_j(Th) \times t_j \geq Th_i,$
  - Constraint #2: Accuracy $\frac{\sum_{j=1}^{n} \mathcal{M}_j(Th) \times t_j \times \mathcal{M}_j(Acc)}{\sum_{j=1}^{n} \mathcal{M}_j(Th)} \geq Acc_i,$
  - Constraint #3: Deadline $\sum_{j=1}^{n} t_j \leq D_i.$
- Among the candidates, choose those with minimal total execution time (to maximize slack).
- Generate exec. plan from the selected models.

Research Thrust #1

- Edge Platform: Nvidia Jetson Xavier
  - 6-core CPUs, 8G RAM, Volta GPUs
- Augmented with multiple TPU acc.
  - 4TOPS @ 2W
- AI Applications and Models

Jetson Xavier

Coral
TPU Accelerator

3 Diff Apps

AI Application	Edge Acceler	Pre-trained DL Models	Size (MB)	Num. Layers
Image Classification	GPU	EfficientNet [36]	5.4	5
		Inception-v1 [34]	6.4	22
		MobileNet-v2 [29]	3.4	20
		SqueezeNet [16]	1.3	15
	TPU	EfficientNet	6.8	7
		Inception-v1	7.0	22
		Inception-v3 [35]	12.0	48
		MobileNet-v1 [15]	1.6	28
Object Detection	GPU	EfficientNet	4.5	5
		MobileNet-v2	6.5	20
		YOLOX [12]	5.4	11
	TPU	EfficientNet	7.6	18
		MobileNet-v1	7.0	28
		MobileNet-v2	6.6	20
Pose Estimation	GPU	MobileNet-v1 [15]	8.2	28
		MoveNet [3]	5.6	25
		PoseNet [25]	4.3	23
	TPU	MoveNet	7.5	25
		PoseNet-MobileNet [25]	1.5	23
		PoseNet-ResNet [25]	24.4	18

GPU Models on TensorFlow

TPU Models on TFLite

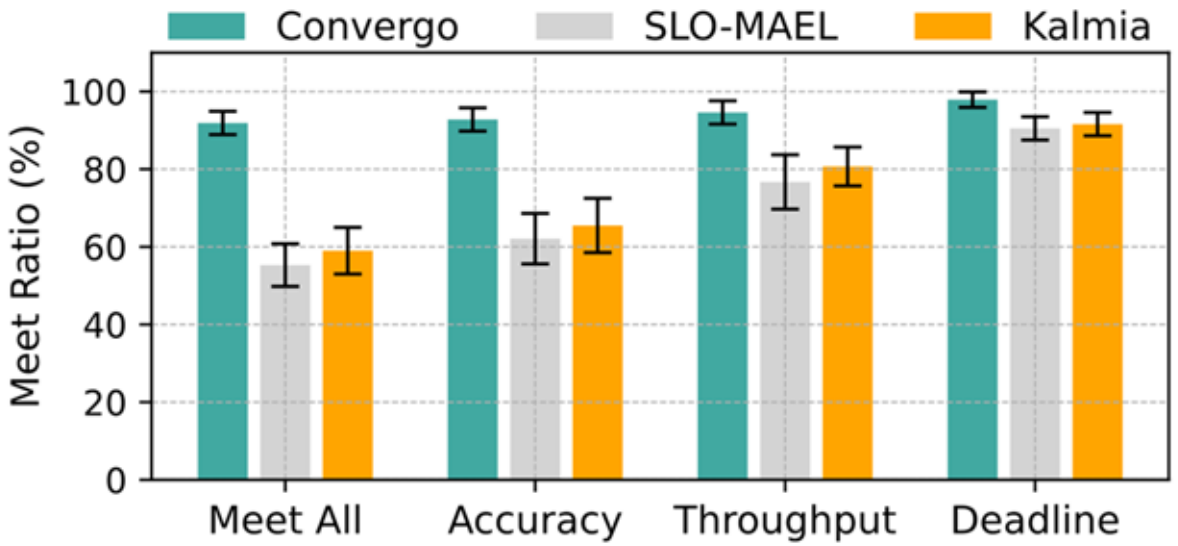
Research Thrust #1

- Workload Generation
  - Mixed AI tasks: IC, OD, PE
  - 500 reqs per workload, generated using Poisson distribution
  - Each req includes unique SLOs: accuracy, throughput, and deadline
  - Evaluated over 10 workload sets to reduce bias and ensure robustness

- Baselines
  - SLO-MAEL (ACM TACO'21): Prioritizes based on estimated latency and risk of SLO violation; uses preemptive execution
  - Kalmia (INFOCOM'24): QoS-aware scheduler for heterogeneous DNNs; combines preemptive and context-aware policies

Research Thrust #1 – Preliminary Results

- Measured both “Meet All” and individual SLOs
  - Meet All: satisfying all three SLOs simultaneously

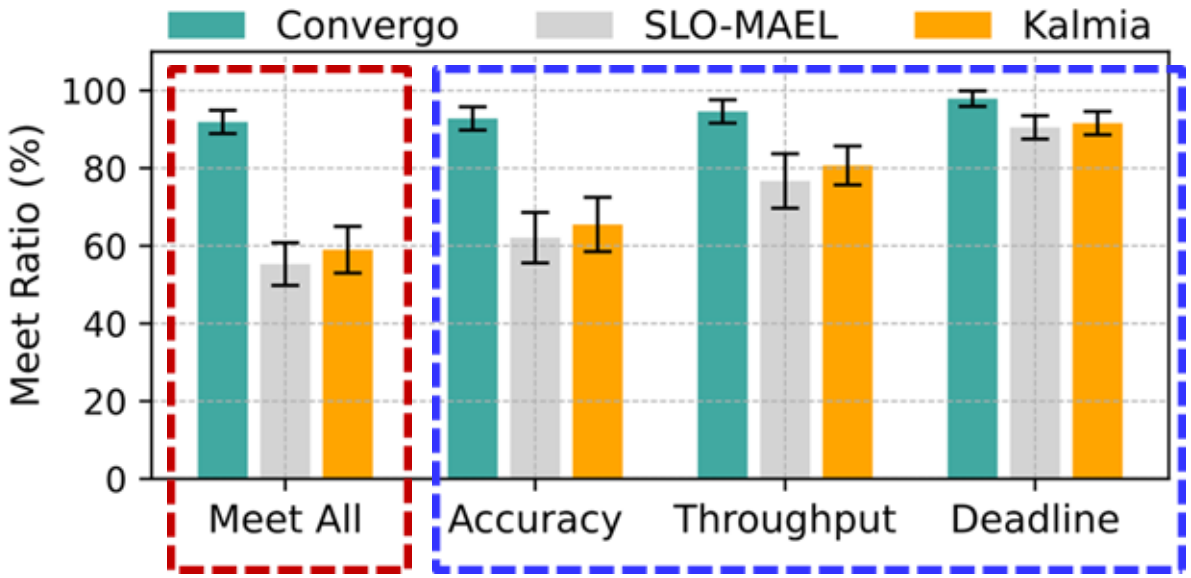


Research Thrust #1 – Preliminary Results

- Measured both “Meet All” and individual SLOs
  - Meet All: satisfying all three SLOs simultaneously

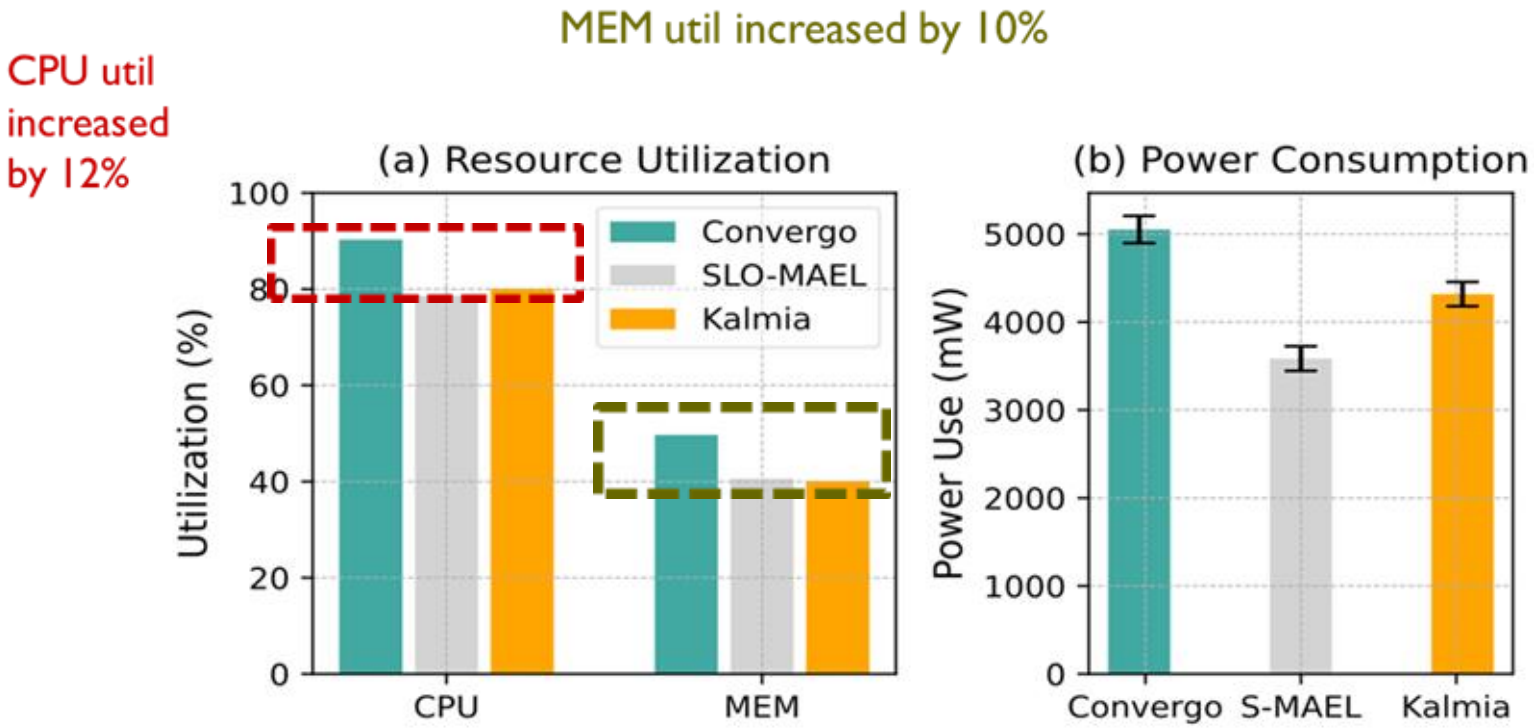
Meet All:
- Convergo: over 95%
- Baselines: less than 60%

Individual SLOs:
- Convergo: 93% (acc), 95% (th), 98% (DL)
- Baselines: less than 70% (acc), less than 80% (th)



Research Thrust #1 – Preliminary Results

- Resource Utilization



Research Thrust #1 – Preliminary Results

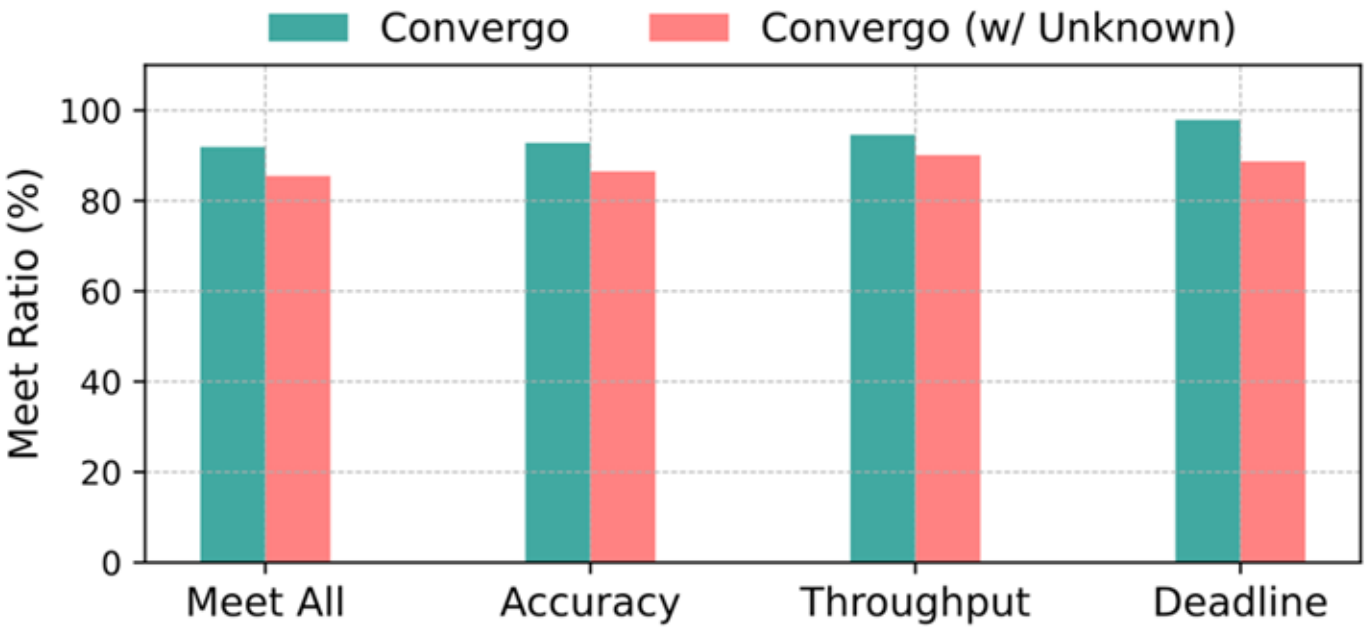
- Covergo Scheduler Overhead

	Avg	Std Dev
Extra CPU Consme..	6.3%	1.6
Extra MEM Consume.	4.9%	1.1
Extra Power Consume.	527mW	45

Still, less than 10% of dev power.

Research Thrust #1 – Preliminary Results

- Performance with ‘unknown’ Models
  - 30% of models were previously unknown (i.e., unprofiled).



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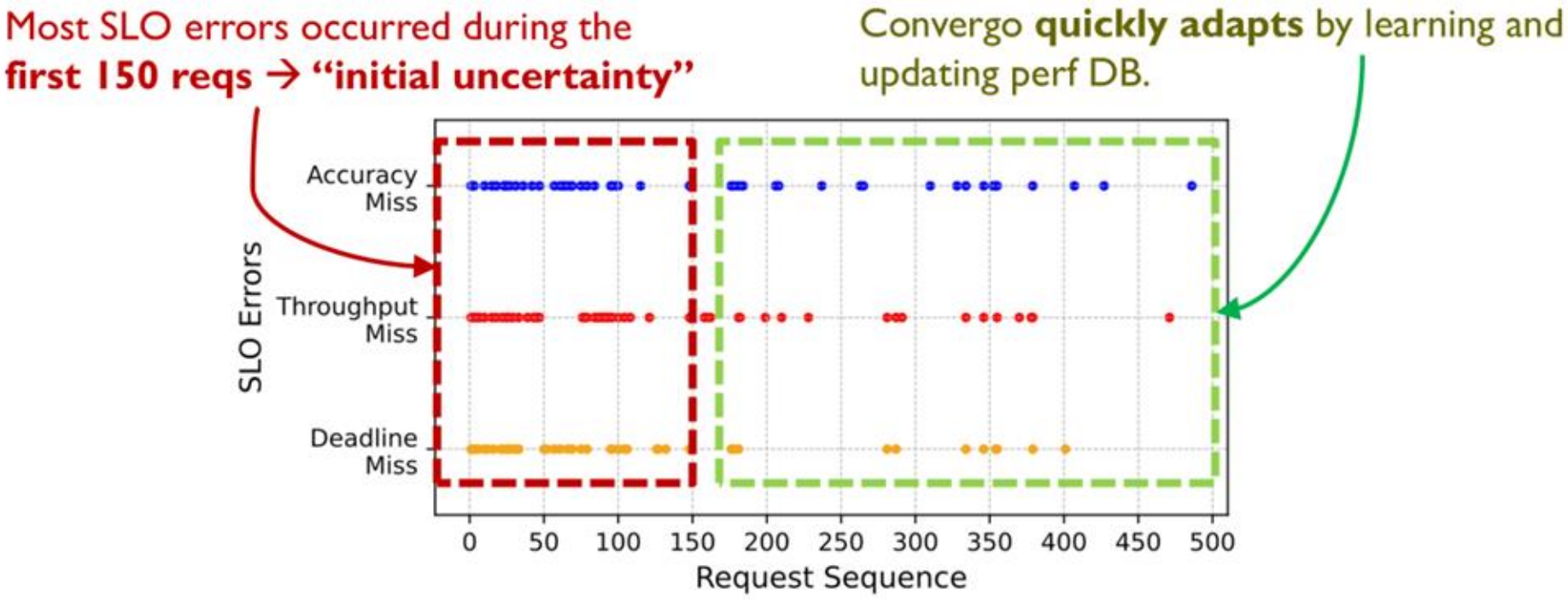


Research Thrust #2:

Model-aware dynamic inference adaptation

Research Thrust #2

- Research goal

  Develop skip-aware models and lightweight prediction mechanisms that expose a quantifiable accuracy performance trade-off space for safe and efficient runtime adaptation.

Research Thrust #2

- Challenge

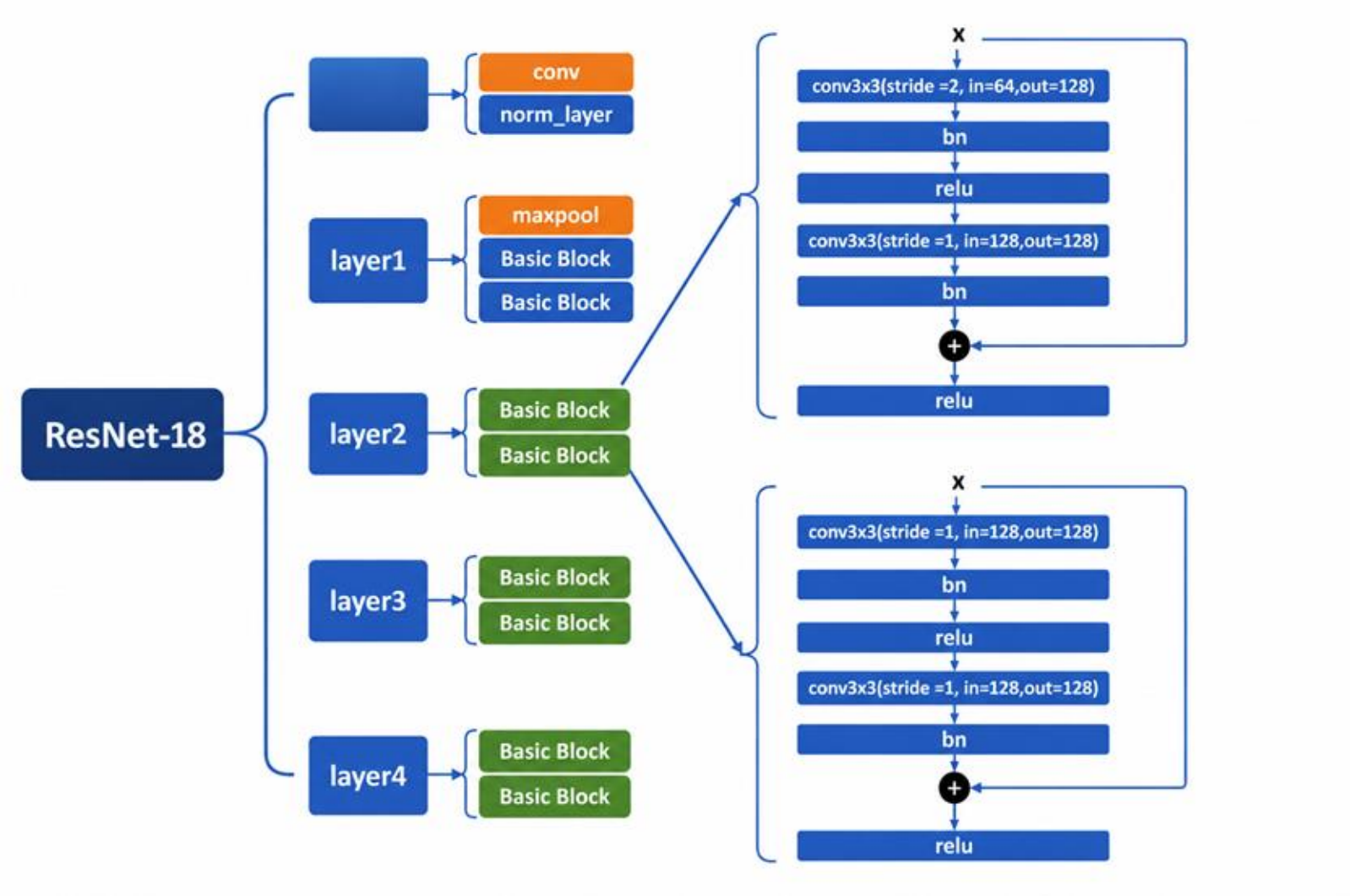
  Real-time Multi-Objective Adaptation under Dynamic Runtime Conditions

  - Configuration effects are highly coupled:
  - Runtime conditions are non-stationary:
  - Decisions must be made online:
  - Existing optimizers (e.g., BO) are too slow for per-inference adaptation

Research Thrust #2

- Bayesian Optimization (BO) Is Not Enough
  - BO assumes:
    - expensive but infrequent evaluations
    - stationary objective function
    - slow convergence acceptable
  - Our system:
    - must decide in milliseconds
    - under changing workload conditions
    - with coupled objectives
    - across unseen skip patterns

Research Thrust #2



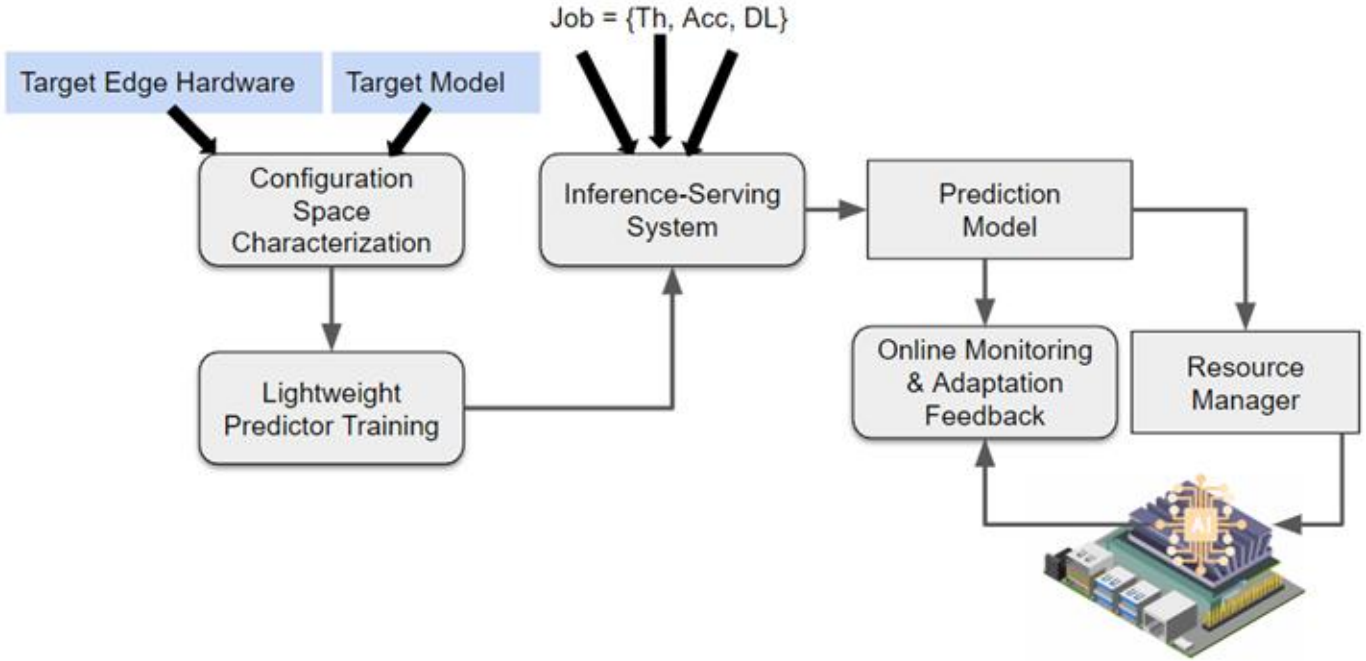
Research Thrust #2

- Offline model

  Enumerates a set of allowable skip configurations

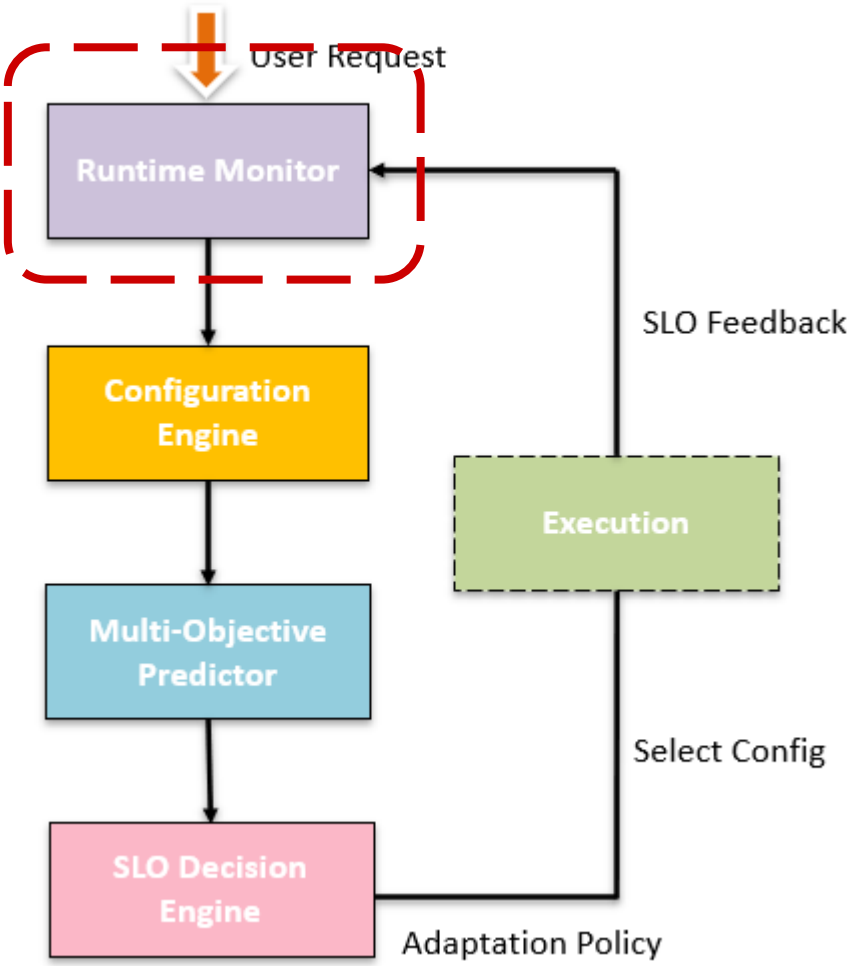
- Online stage

  Select a suitable skip configuration



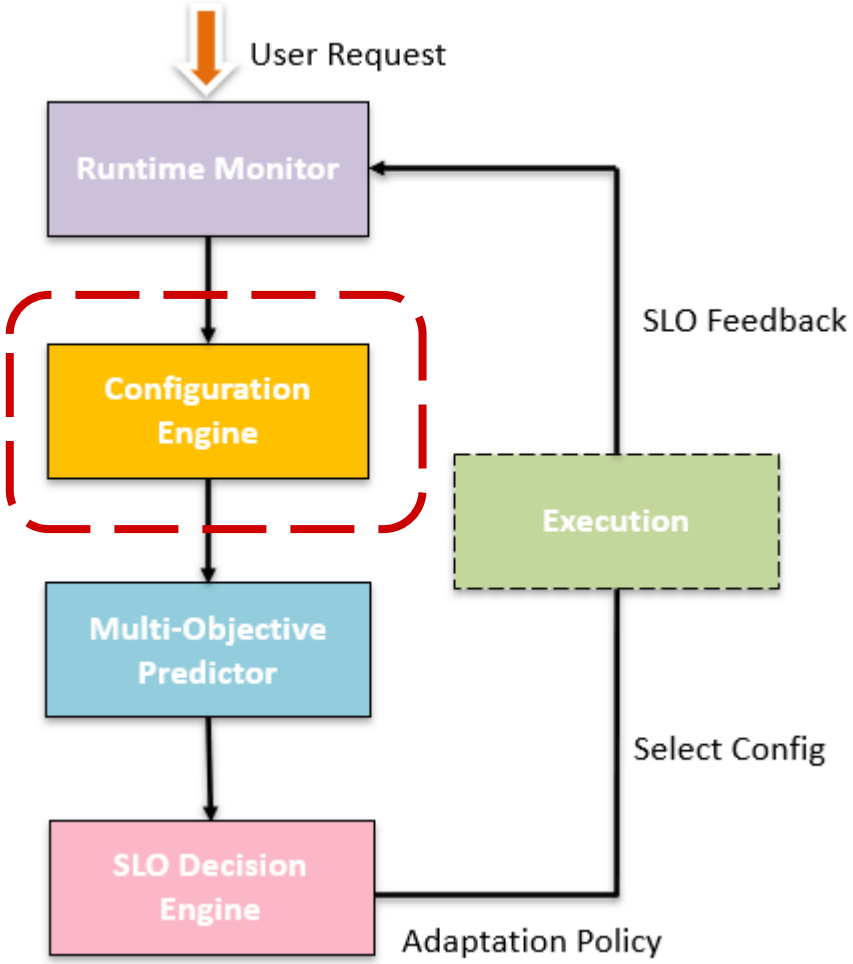
Research Thrust #2

- Runtime Monitor
  - Power State
  - Device Load
  - Queue Length



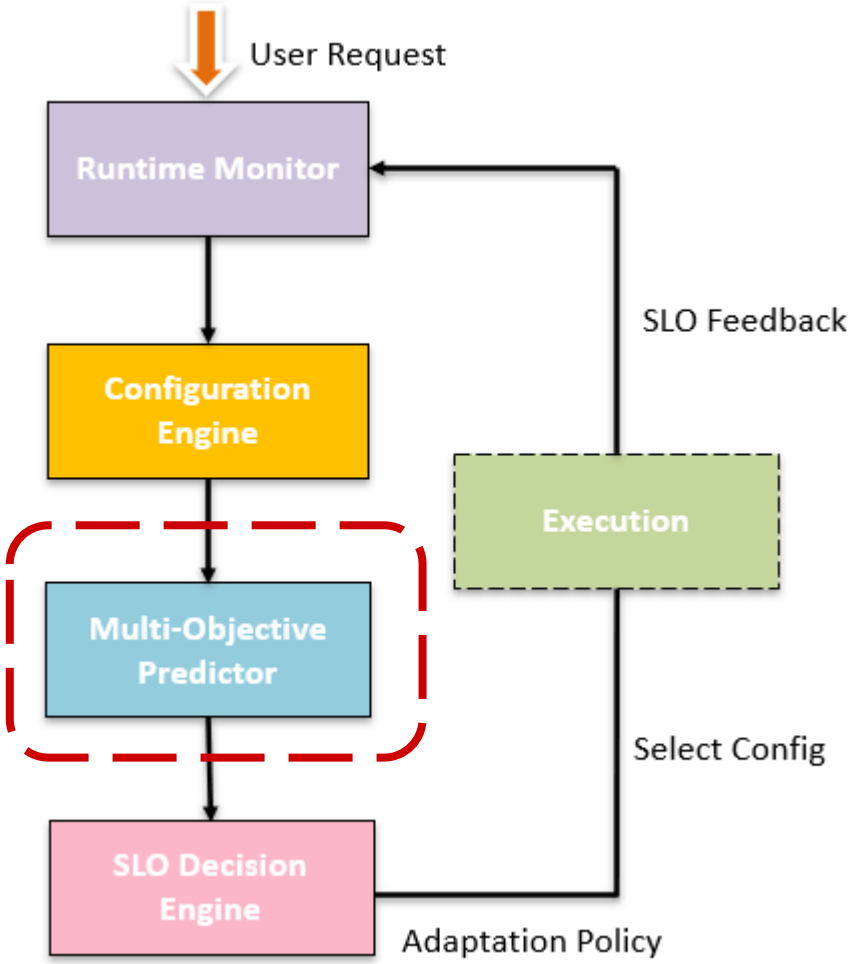
Research Thrust #2

- Configuration Engine
  - Skip Config
  - Batch Size
  - Power Mode



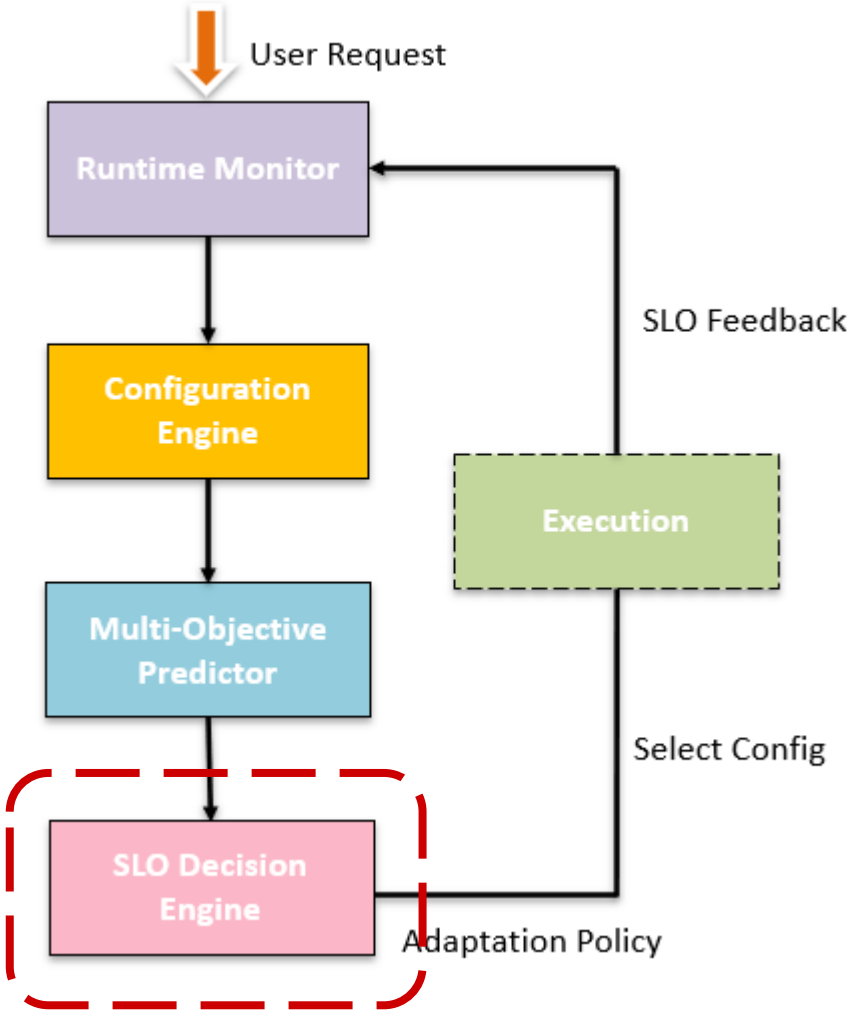
Research Thrust #2

- Multi-Objective Predictor
  - Accuracy Predictor
  - Latency Predictor
  - Energy Predictor



Research Thrust #2

- SLO Decision Engine
  - Accuracy Constraint
  - Deadline Constraint
  - Energy Objective

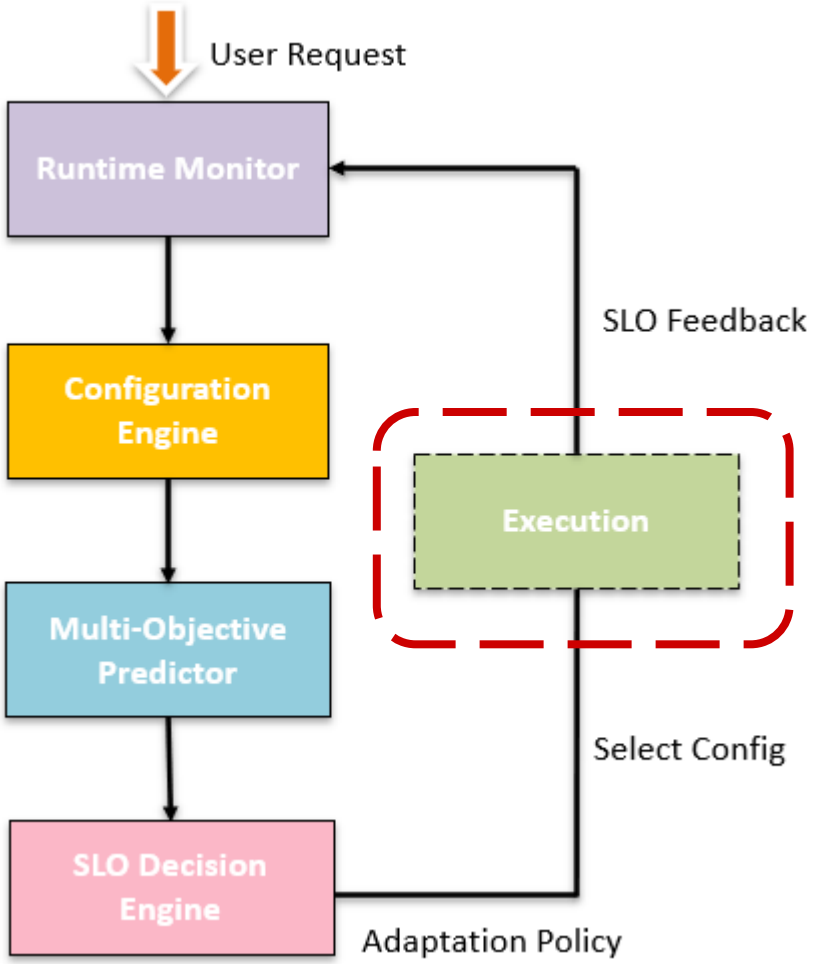


Research Thrust #2

- Execution

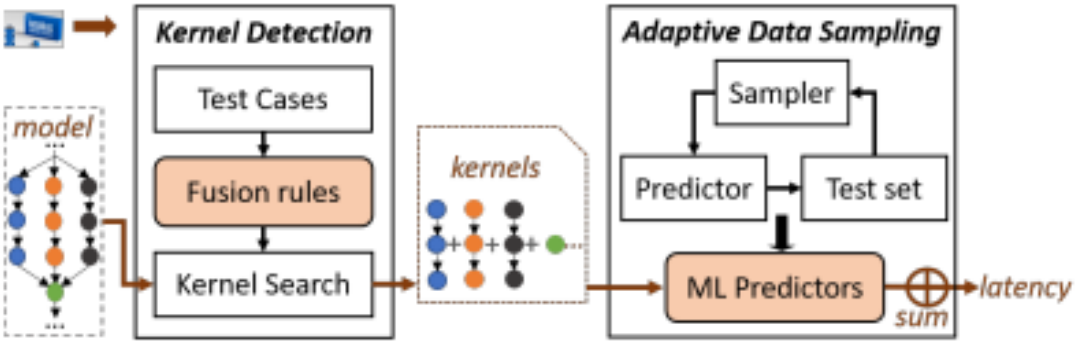
  Execution on heterogeneous devices

  - Jetson Orin Nano
  - Jetson AGX Xavier
  - GPU



Research Thrust #2 -- Related Work

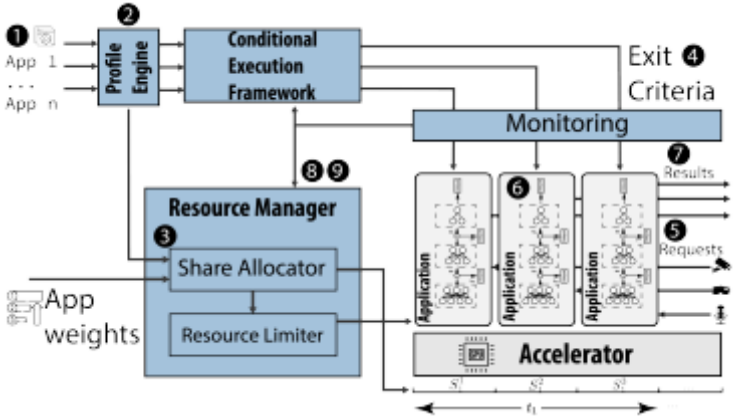
- Zhang, Li Lyna, et al. (MobiSys'21)
  - Proposes a learning-based framework to predict DNN inference latency on diverse edge devices.
  - Builds a latency predictor using operator-level profiling and regression modeling.



- Limitations
  - Predicts latency only; does not model accuracy, energy consumption, or multi-objective trade-offs needed for SLO satisfaction.
  - Requires extensive offline kernel profiling for each new device; adaptation overhead limits real-time reconfiguration scenarios.
  - Static prediction model cannot adapt to dynamic system conditions

Research Thrust #2 -- Related Work

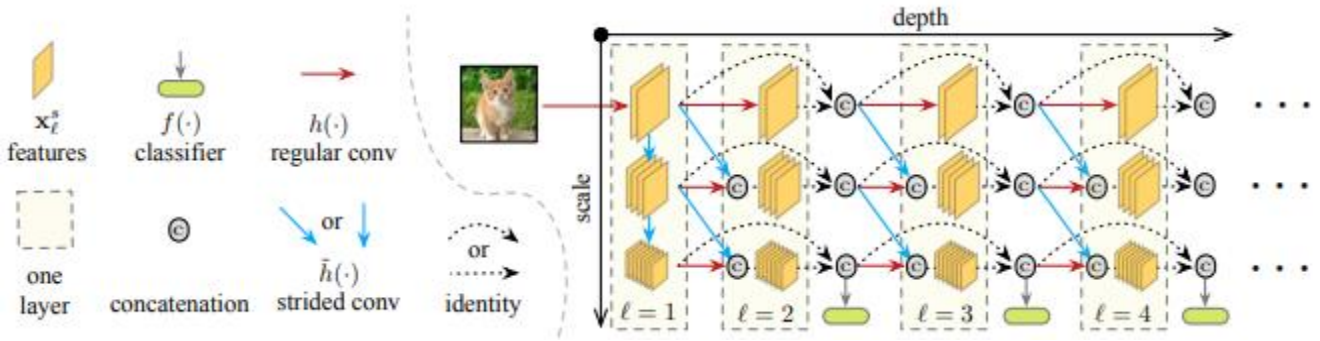
- Liang, Qianlin, et al. (IoTDI'23)
  - Adaptive model-serving system for multi-tenant edge environments that dynamically adjusts batch size and power modes.
  - Uses profiling-based performance models to predict latency and resource usage for different configurations.



- Limitations
  - Treats models as static black-box variants (no internal structural elasticity).
  - Adaptation occurs at the model granularity, not layer/block level.
  - Does not provide predictable accuracy–performance degradation modeling.

Research Thrust #2 -- Related Work

- Huang et al. (ICLR 2018)
  - Introduces multi-scale dense networks with multiple early-exit classifiers.
  - Enables adaptive inference by terminating computation once sufficient confidence is reached.
  - Provides accuracy-latency tradeoffs for resource-constrained deployment



- Limitations
  - Adaptation is driven by confidence thresholds rather than explicit SLO objectives.
  - Does not jointly optimize latency, energy, and accuracy.
  - Exit policies are largely fixed after training and do not adapt to runtime system states.
  - Lacks workload-aware and device-aware online optimization.

Research Thrust #2 -- Related Work

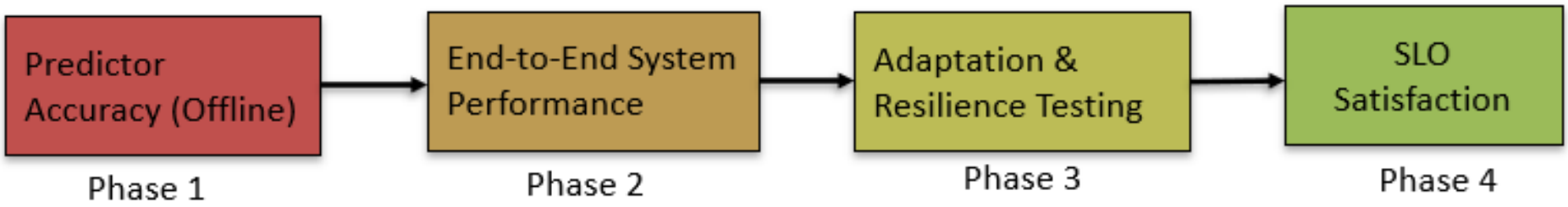
- Wang et al. (ECCV 2018)
  - Introduces dynamic layer-skipping through learnable gating networks.
  - Routes each input through different residual blocks based on input difficulty.
  - Reduces infere execution.

- Limitations
  - Optimizes inference efficiency only; does not consider energy-aware multi-SLO objectives.
  - Decisions depend primarily on input features, not runtime system conditions.
  - Assumes a fixed deployment environment without workload-aware adaptation.
  - Does not explicitly model accuracy–latency–energy tradeoff surfaces for runtime optimization.

Evaluation Plan

- RQ1: How accurate are our lightweight prediction models (latency + accuracy)?
- RQ2: Can dynamic layer-skipping meet diverse SLOs while reducing energy?
- RQ3: What is the overhead of runtime decision-making?
- RQ4: How does the system adapt to workload variations?

Evaluation Plan



Evaluation Plan

- Hardware Platforms:

Device	CPU	GPU	Memory	Power	Price
Jetson Orin Nano	6-core Arm Cortex-A78AE	1024-core NVIDIA Ampere	8GB	7–15W	$499
Jetson AGX Xavier	8-core Carmel ARM	512-core Volta + Tensor	32GB	10–30W	$699
GeForce RTX 2080 Ti	8-core 64-bit CPU	4352 CUDA cores	16GB	250W	$999
RTX A5000	48-core 64-bit CPU	8192 CUDA cores	196GB	230W	$2500

Evaluation Plan

- Models:

  ResNet-18, ResNet-34, ResNet-50, ResNet-74, ResNet-101, ResNet-152

- Dataset: CIFAR-10 and CIFAR-100
- Layer-skip configurations: 4,095 paths per model (ResNet-34 example)
- Batch sizes: {1, 2, 4, 8, 16, 20}
- Power modes: 8 combinations (4 modes × jetson_clocks on/off)
- Total: About 160,000 unique configurations per model

Evaluation Plan

Baselines:

- B1: nn-Meter (MobiSys'21)
  - Builds device-specific performance models using operator-level profiling
  - Predicts latency only
- B2: Dělen (IoTDI'23)
  - Dynamically adjusts batch size and power modes
  - Limited multi-objective optimization
- B3: MSDNet (ICLR'18)
  - Multi-exit neural architecture with adaptive early termination
  - Not designed for multi-SLO scheduling
- B4: SkipNet (ECCV'18)
  - Dynamically skips residual blocks using learned gating networks
  - Optimizes model execution only

Evaluation Plan

Comparison Matrix:

System	Runtime Adaptive	Accuracy-Aware	Energy-Aware	Multi-SLO
nn-Meter	✗	✗	✗	✗
Dělen	✓	Partial	Partial	Partial
MSDNet	Partial	✓	✗	✗
SkipNet	✓	✓	✗	✗
Thrust 2	✓	✓	✓	✓

Evaluation Plan

Expected Results:

- Prediction Accuracy
- SLO Satisfaction
- Energy Savings
- Overhead
  - Prediction time
  - Total overhead

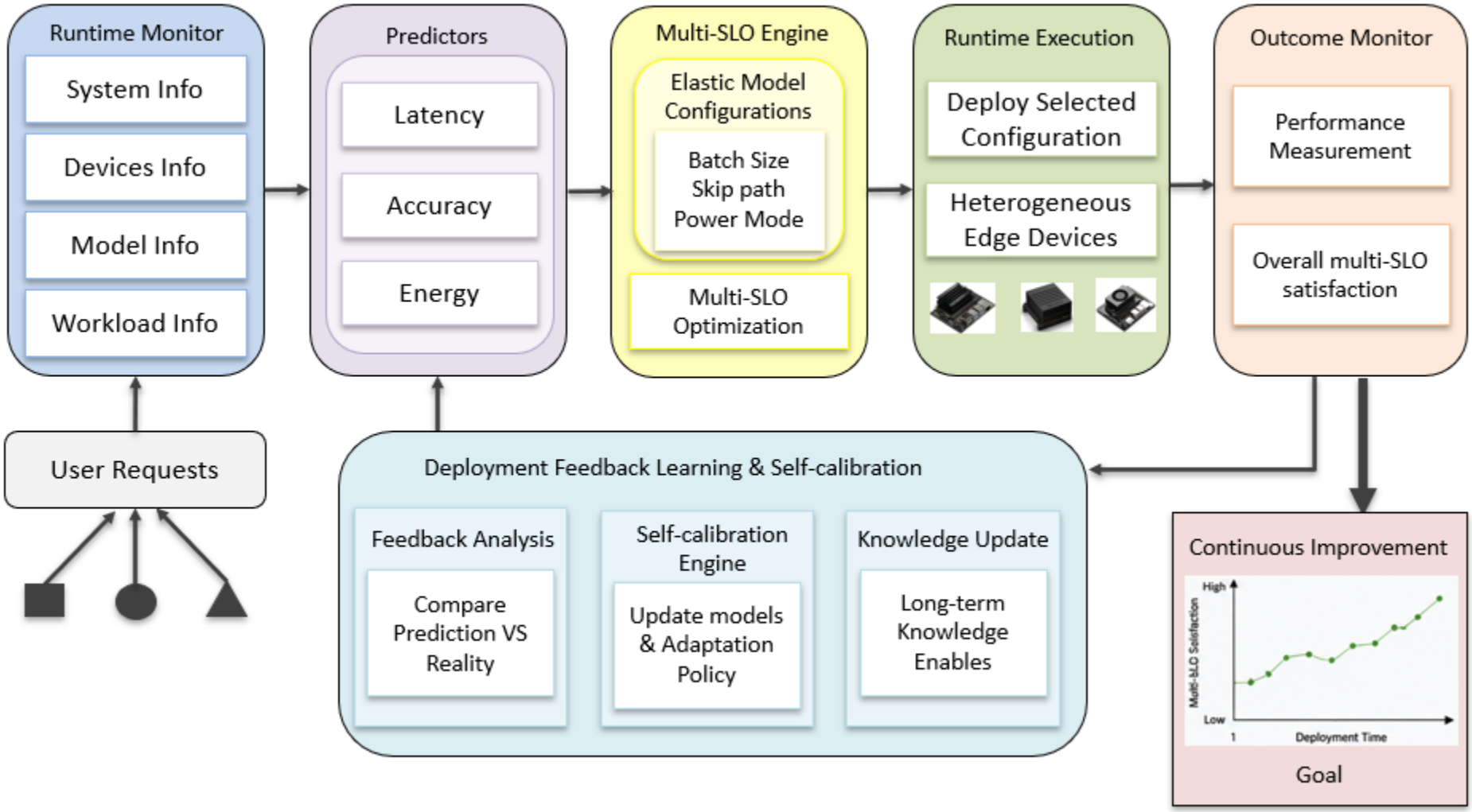
Research Thrust #3:

Self-Calibrating Multi-SLO Adaptation

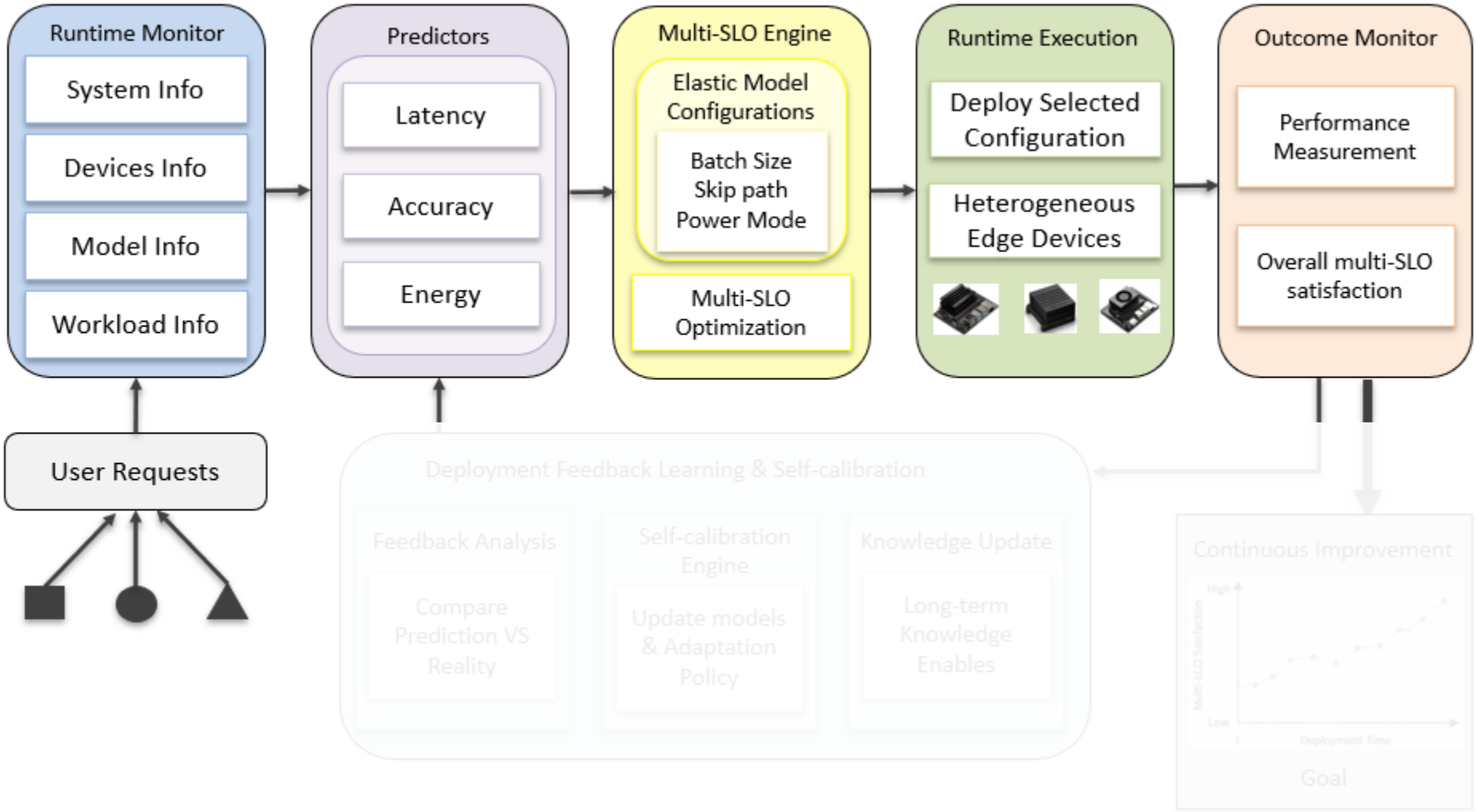
Research Thrust #3

- Challenge
  - Runtime environments are non-stationary
  - Static adaptation policies cannot learn from experience
  - Prediction models gradually become stale
- Research goal
  - Develop deployment-feedback mechanisms that continuously calibrate adaptation models under evolving workload and device conditions.

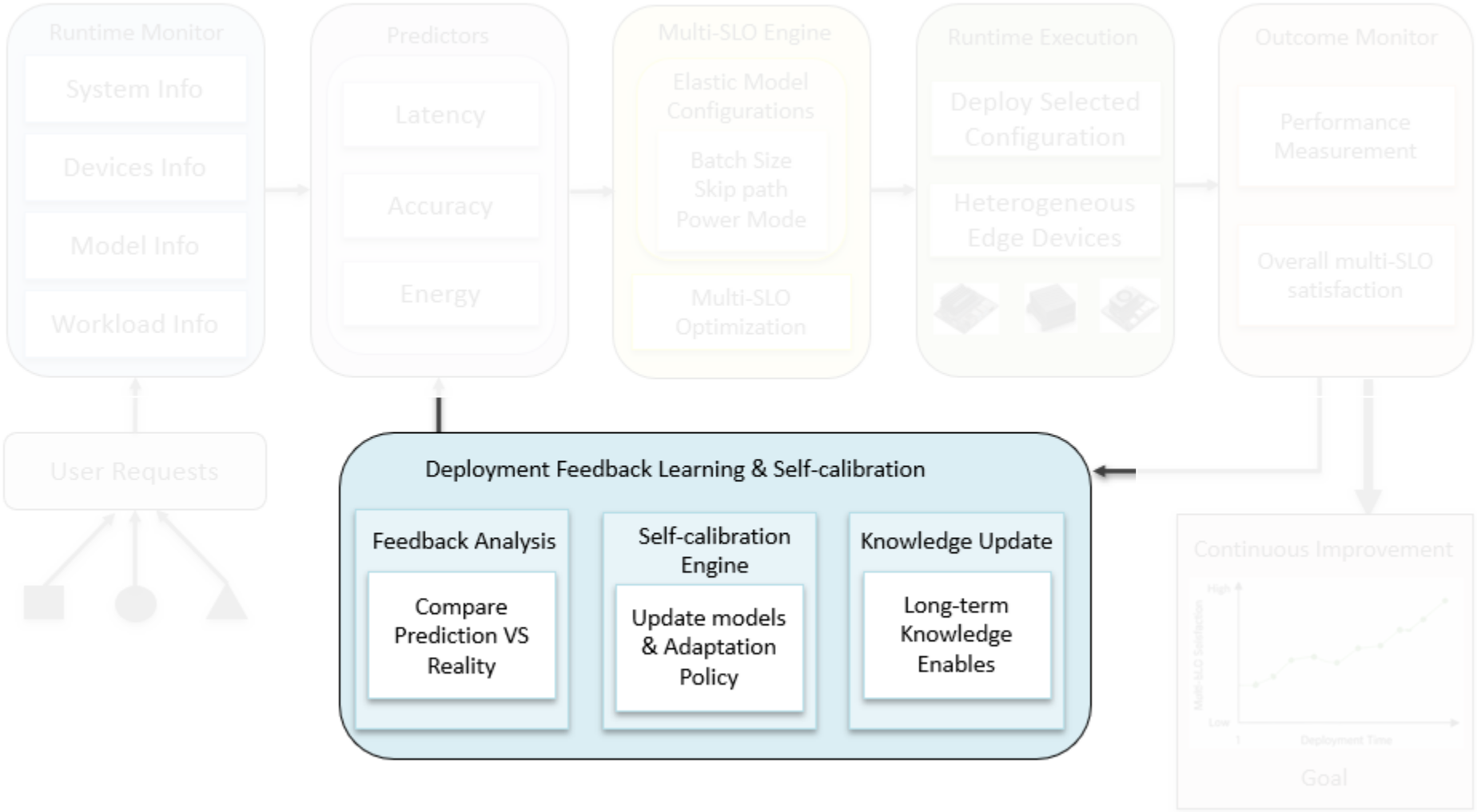
Research Thrust #3



Research Thrust #3



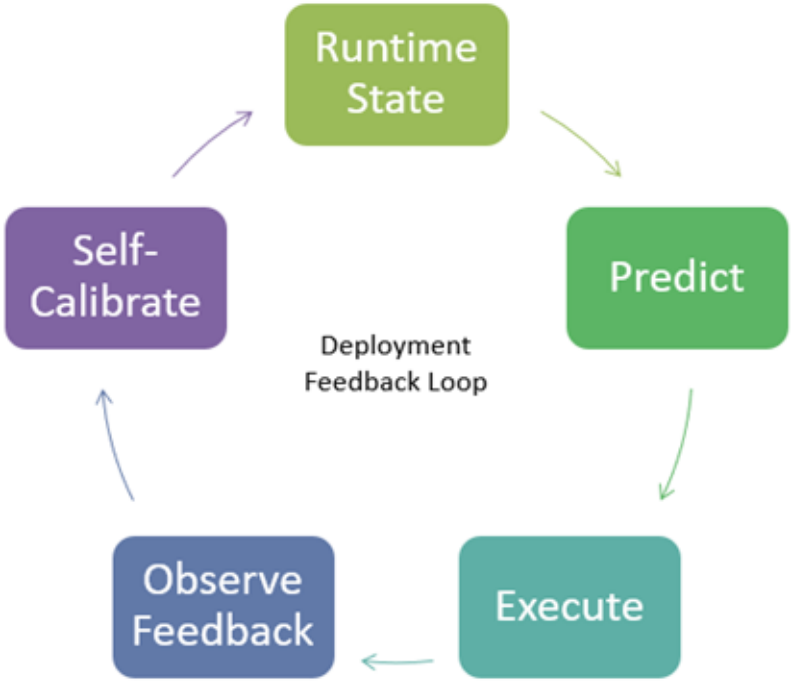
Research Thrust #3



Research Thrust #3

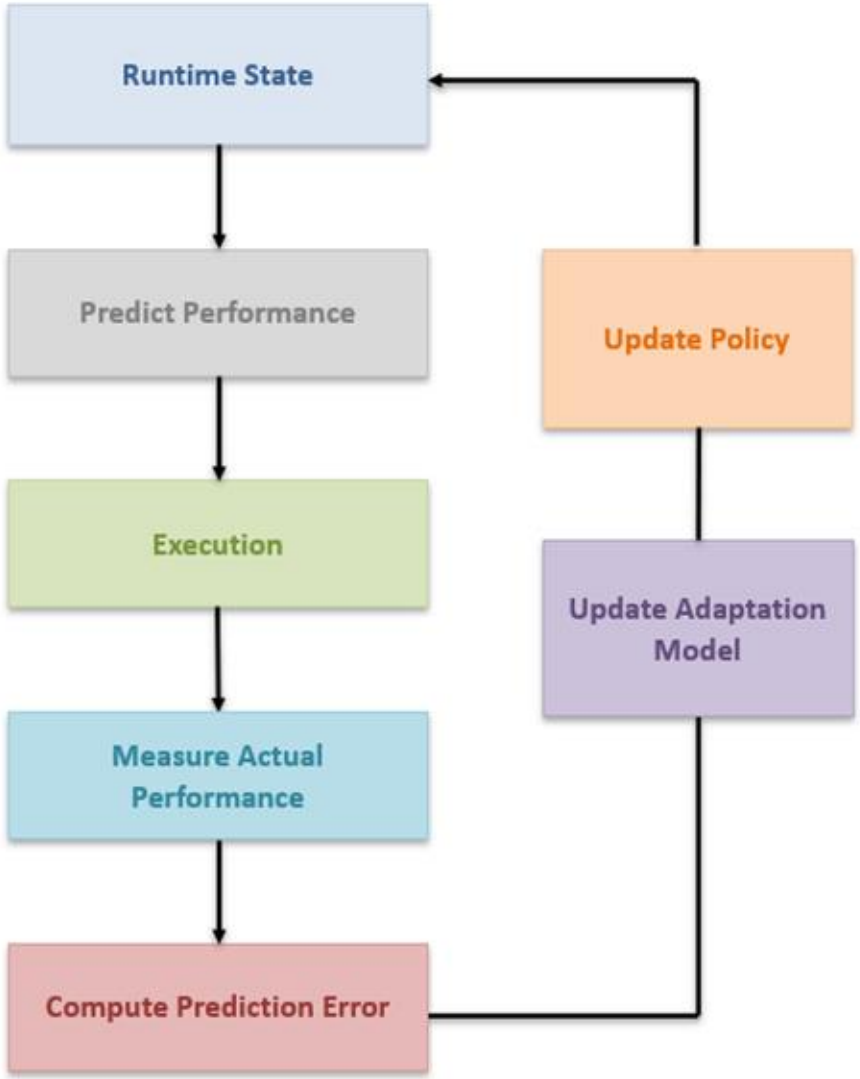
Self-calibration Runtime Framework

- Self-Calibration loop that continuously learns from deployment feedback and maintains adaptation effectiveness over time.



Research Thrust #3

- Deployment Feedback and Self-calibration



Research Thrust #3 -- Related Work

- Wang, L. et al. (ICASSP' 24)
  - Uses adaptive multi-armed bandits for runtime decision-making.
  - Learns from historical execution outcomes.
  - Continuously updates decisions based on observed rewards.
- Limitations
  - Focuses on task offloading decisions.
  - Does not optimize multi-SLO satisfaction.

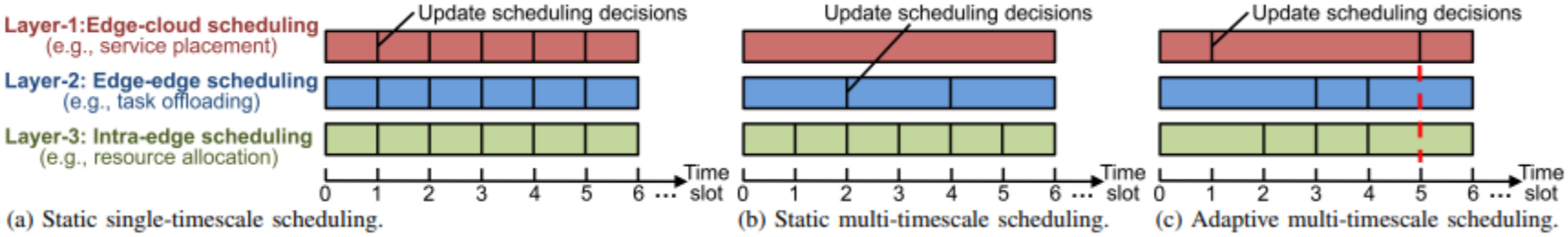
Algorithm 1 Adaptive Task Offloading Algorithm

Initialization: ε, ξ; $\mathcal{A}=\{\mathbf{A_1}, ..., \mathbf{A_N}\}$; $N_{\mathbf{A_n}} = 0$.
Exploration:
for $t = 1, \ldots, S$ **do**
  Choose an arm $\mathbf{A_n}$ from arm space $\mathcal{A}$;
  Calculate $R_{\mathbf{A_t}}(t)$;
  Update memory pool with current network traffic state;
end for
Exploration-Exploitation:
for $t = S+1, \ldots, T$ **do**
  Predict network state s_t according to variance v_t;
  if $s_t = 1$ **then**
    Adopt ε-greedy method:
    Generate a random number $randnum$;
    if $randnum < \varepsilon$ **then**
      Choose an arm $\mathbf{A_n}$ from $\mathcal{A} \backslash \mathcal{A}'$ randomly;
      Add the chosen arm $\mathbf{A_n}$ into $\mathcal{A}'$;
      Calculate $R_{\mathbf{A_n}}(t)$;
    else
      Choose optimal arm $\mathbf{A_{opt}} = \arg\min_{\mathbf{A_n} \in \mathcal{A}'} R_{\mathbf{A_n}}(t)$;
      Calculate $R_{\mathbf{A_{opt}}}(t)$;
    end if
  else
    Adopt UCB1-based method:
    for $n = 1, \ldots, N$ **do**
      Estimate expected reward $\hat{R}_{\mathbf{A_n}}(t)$ in (8);
    end for
    Choose the optimal arm $\mathbf{A_{opt}} = \arg\min_{\mathbf{A_n} \in \mathcal{A}} \hat{R}_{\mathbf{A_n}}(t)$;
    Calculate $R_{\mathbf{A_{opt}}}(t)$;
    $N_{\mathbf{A_n}} = N_{\mathbf{A_n}} + 1$;
  end if
  Update cumulative average reward and memory pool;
end for

Research Thrust #3 -- Related Work

- Hao, Y. et al. (ICASSP' 24)
  - Proposes a deep reinforcement learning framework for adaptive scheduling in mobile edge computing environments.
  - Learns scheduling policies across multiple time scales to handle dynamic workloads and resource variations.
  - Continuously adapts scheduling decisions based on observed system feedback.

- Limitations
  - Assumes fixed application models and does not consider internal model configurations.
  - Does not optimize accuracy, latency, and energy jointly



Research Thrust #3 -- Related Work

- Wirth, M. et al. (GLOBECOM ' 23)
  - Uses contextual multi-armed bandits to make runtime decisions in non-stationary edge environments.
  - Learns from historical observations and continuously updates decision policies.
- Limitations
  - Focuses on server selection and task offloading decisions.
  - Does not consider elastic DNN configuration spaces.
  - Does not address multi-SLO optimization.

Algorithm 1 C-CPD-UCB

Input Parameters: T, W, ξ and δ
Set $N_{k,0}(\alpha) = 0, \quad \forall k, \forall \alpha$.
Set $\tau = 0$.
for each $t = 1, \ldots, T$ **do**
  Observe task type α_t.
  if $N_{k,t-1}(\alpha_t) > 0 \; \forall k \in \mathcal{K}$ **then**
    Select BS $\hat{k}_t$ according to (10).
  else
    Select BS $\hat{k}_t$ randomly from set $\{k \in \mathcal{K} | N_{k,t-1}(\alpha_t) = 0\}$.
  end if
  Observe delay $T_{\hat{k}_t,t}(\alpha_t)$ and update $\hat{\mu}_{\hat{k}_t,t}(\alpha_t)$, $N_{\hat{k}_t,t}(\alpha_t)$ and $c_{\hat{k}_t,t}(\alpha_t)$ based on Equations (11), (13) and (12), respectively.
  if Equation (14) is satisfied **AND** $N_{k,t}(\alpha) > 0 \; \forall k \in \mathcal{K}, \forall \alpha \in \mathcal{A}$ **then**
    Set $\tau = t - W$.
    for $k \in \mathcal{K}$ **do**
      for $\alpha \in \mathcal{A}$ **do**
        Update $N_{k,t}(\alpha)$ according to Equation (13).
        if $N_{k,t}(\alpha) > 0$ **then**
          Update $\hat{\mu}_{k,t}(\alpha)$ and $c_{k,t}(\alpha)$ based on Equations (11) and (12), respectively.
        end if
      end for
    end for
  end if
end for

Evaluation Plan

- Set up

  Edge Devices: Jetson Orin Nano, Jetson AGX Xavier, RTX 2080 Ti, RTX A5000

  Learning Framework: Contextual Bandit / History-Aware Policy

  Predictor Models: Accuracy, Latency, Energy

- Metric

  Multi-SLO Satisfaction Rate

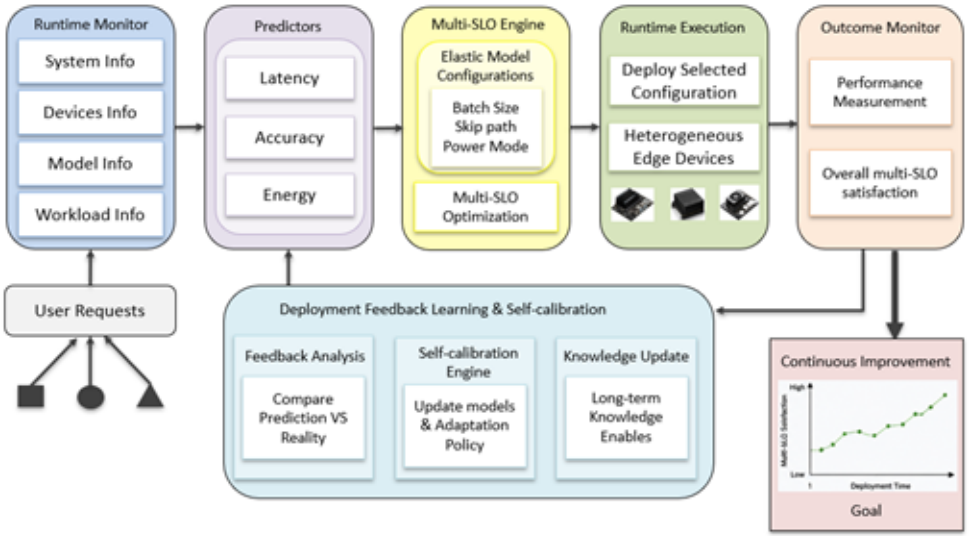
  Calibration Effectiveness

  Runtime Overhead

- Baselines

  EdgeTimer

  Contextual Multi-Armed Bandits (CMAB)



Agenda

- ❑ Motivation and Research Challenges.
- ❑ Proposed Approach
  1. Research Part #1 – Multi-SLO heterogeneous scheduling
  2. Research Part #2 – Model-aware dynamic inference adaptation
  3. Research Part #3 – Self-Calibrating Multi-SLO Adaptation
  4. Evaluation Plan
- ❑ **Timeline, Summary, and Contributions**

Timeline

	Semester:	1	2	3	4	5
Research	Thrust 1	■	■			
	Thrust 2		■	■	■	
	Thrust 3				■	■
Paper	Paper Submission		■			■
Thesis	Thesis Writing & Defense					■

Table 2: **Project Timeline**

Summary and Contributions

- Summary
  - This project advances edge AI from static scheduling, to predictor-guided adaptation, and ultimately to self-calibrating deployment-feedback-driven intelligence for sustained multi-SLO satisfaction.
- Research Contributions
  - Multi-SLO scheduling across heterogeneous edge devices
  - Prediction-based exploration of elastic configuration spaces
  - self-calibrate adaptation using deployment feedback
  - Robust multi-SLO satisfaction under dynamic environments

Thank you!